

Given: $x = (5, 10, 25)$, $B = \left\{ \left(\frac{3}{5}, \frac{4}{5}, 0 \right), \left(-\frac{4}{5}, \frac{3}{5}, 0 \right), (0, 0, 1) \right\}$

$$[x]_B = \begin{bmatrix} x \cdot B_1 \\ x \cdot B_2 \\ x \cdot B_3 \end{bmatrix} \quad \text{Since } B = \{B_1, B_2, B_3\} \text{ is orthonormal.}$$

$$1.) x \cdot B_1 = (5, 10, 25) \cdot \left(\frac{3}{5}, \frac{4}{5}, 0 \right) = 5\left(\frac{3}{5}\right) + 10\left(\frac{4}{5}\right) + 0 = 3 + 8 + 0 = 11$$

$$2.) x \cdot B_2 = (5, 10, 25) \cdot \left(-\frac{4}{5}, \frac{3}{5}, 0 \right) = 5\left(-\frac{4}{5}\right) + 10\left(\frac{3}{5}\right) + 0 = -4 + 6 + 0 = 2$$

$$3.) x \cdot B_3 = (5, 10, 25) \cdot (0, 0, 1) = 0 + 0 + 25(1) = 25$$

$$\text{Thus, } [x]_B = \begin{bmatrix} 11 \\ 2 \\ 25 \end{bmatrix}$$